To completely eliminate architectural bias in your comparison, the absolute best strategy is to select an SLM and an LLM from the **exact same model family**.

When researchers at the same lab release a "suite" of models, they typically use the exact same vision encoder, the exact same training data pipeline, and the exact same core attention mechanisms. The *only* major variable that changes is the parameter count of the text backbone. This isolates the "scale vs. data" argument perfectly for your project.

Here are the three best open-source, multimodal model pairs for your video contextualization project:

Pair 1: The Qwen2.5-VL Family (Alibaba Cloud)

This is arguably the strongest open-source multimodal family currently available for video processing. They natively support dynamic resolution and long-form video tokenization.

- **The LLM:** Qwen2.5-VL-72B (72 Billion parameters)
- **The SLM:** Qwen2.5-VL-3B or Qwen2.5-VL-7B
- **Why the bias is minimized:** Both models use the exact same ViT (Vision Transformer) architecture and the same unique "Multimodal Rotary Positional Embedding" (mRoPE) designed specifically for 1D time, 2D spatial, and 3D video data. If the 72B model beats the 3B model, you can mathematically prove it was due to the parameter scale, not because it had a better "eye" for the video.

Pair 2: The Llama 3.2 Vision Family (Meta)

Meta released these models simultaneously, making them perfect for a controlled academic study.

- **The LLM:** Llama-3.2-90B-Vision
- **The SLM:** Llama-3.2-11B-Vision
- **Why the bias is minimized:** Unlike models that fuse vision and text early on, Llama 3.2 uses a very specific adapter-based cross-attention architecture. Both the 90B and 11B models were trained on the exact same high-quality data mixture using the exact same image-text alignment techniques. Comparing them provides a 1-to-1 look at how scaling the underlying Llama 3 text parameters impacts video reasoning.

Pair 3: The InternVL 2.5 Family (OpenGVLab)

InternVL is highly respected in the academic community for its transparency and open-weight ecosystem.

- **The LLM:** InternVL2.5-78B
- **The SLM:** InternVL2.5-8B
- **Why the bias is minimized:** Both models utilize the same dynamic high-resolution

patching strategy (allowing them to process videos as a series of high-res image tiles). They share identical alignment strategies and instruction-tuning datasets.

How to Mitigate the Remaining Bias (Minimal Effort)

Even within the same family, you must enforce a strict testing environment in your code to keep the comparison fair:

1. **Lock the Frame Extraction:** Write a Python script using OpenCV to extract the exact same number of frames (e.g., 1 frame per second) and feed that identical array to both models. Do not let the LLM's larger context window tempt you into giving it more frames.
2. **Standardize Quantization:** Running a 72B model requires massive VRAM. You will likely need to quantize it (e.g., using 4-bit AWQ or GPTQ) to run it locally or on a cloud instance. If you quantize the LLM to 4-bit, you **must** also quantize the SLM to 4-bit to ensure the precision loss is identical across both tests.
3. **Identical System Prompts:** Use the exact same zero-shot or few-shot system prompt for both models. Do not tweak the SLM prompt to "help it out."